

$$26 \div a, \quad 1 = ax + 26y$$

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1. Inverse

$$1) 1 \times 1 \bmod 26 = 1 \quad 3$$

$$\underline{1} \rightarrow 1$$

$$\textcircled{1} \quad 3 \Rightarrow 26 = 3(8) + 2$$

$$3 = 2(1) + 1$$

$$\gcd(3, 26) = 1 \quad \checkmark$$

$$1 = 3 - 2 \rightarrow \textcircled{1}$$

$$2 = 26 - 3(8) \rightarrow \textcircled{2}$$

$$1 = 3 - 26 + 3(8)$$

$$1 = 27 - 26$$

$$1 = 3(9) - 26$$

$$1 = 3(9) + 26(-1)$$

$$\boxed{3^{-1} = 9 \bmod 26}$$

$$\textcircled{2} \quad 5 : 26 = 3(8) + 1$$

$$\underline{5} = 26 - 3(8)$$

$$1 = 26 - 5(5)$$

$$1 = 5(-5) + 26(1)$$

$$5^{-1} = (-5) \bmod 26$$

$$-5 + 26 = 21$$

$$\boxed{5^{-1} = 21}$$

$$\textcircled{3} \quad 7$$

$$26 = 7(3) + 5 \Rightarrow 5 = 26 - 7(3)$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$\text{Back: } 1 = 5 - 2(2) \Rightarrow \textcircled{1}$$

$$2 = 7 - 5 \rightarrow \textcircled{2}$$

$$\textcircled{1} \Rightarrow 1 = 5 - 2(7 - 5)$$

$$1 = 5 - 14 + 10$$

$$1 = 3(5) - 2(7)$$

$$5 = 26 - 7(3) \rightarrow \textcircled{3}$$

$$1 = 3(26 - 7(3)) - 2(7)$$

$$1 = 3(26) - 9(7) - 2(7)$$

$$1 = 3(26) - 11(7)$$

$$\text{coeff of } 7 = -11$$

$$\boxed{7^{-1} = 15}$$

$$\textcircled{4} \quad 9$$

Brute force

$$a a^{-1} \bmod 26 = 1$$

$$3 \times 1 \bmod 26 = 3$$

$$3(2) \bmod 26 = 6$$

$$3(3) \bmod 26 = 9$$

$$3 \times 4 \bmod 26 = 12$$

$$3(6) \bmod 26 = 18$$

$$3(8) \bmod 26 = 24$$

$$3(9) \bmod 26 = 1$$

$$\Rightarrow \boxed{3^{-1} = 9}$$

⑤ 11

$$26 = 11(2) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3(1) \rightarrow \textcircled{1}$$

$$3 = 11 - 4(2) \rightarrow \textcircled{2}$$

$$\textcircled{1} \Rightarrow 1 = 4 - 11 + 4(2) \uparrow$$

$$1 = 4(3) - 11$$

$$\Rightarrow 2 \quad 4 = 26 - 11(2) \rightarrow \textcircled{3}$$

$$1 = 2(26 - 11(2)) - 11(1)$$

$$= 26 - 11(3)$$

$$1 = 26(3) - 6(11) - 1(11)$$

$$1 = 26(3) - 7(11)$$

$$\Rightarrow -11 = 26 - 7$$

$$\Rightarrow \boxed{11^{-1} = 19}$$

⑥ $26 = 15(1) + 11$

$$15 = 11(1) + 4$$

$$11 = 4(2) + 3$$

$$4 = 3(1) + 1$$

$$1 = 4 - 3(1) \rightarrow \textcircled{1}$$

$$\Rightarrow 1 = 4 - 11 + 4(2)$$

$$1 = 4(3) - 11$$

$$1 = 3(15 - 11) - 11$$

$$1 = 15(3) - 11(4)$$

$$1 = 15(3) - 26(4) + 15(4)$$

$$1 = 15(7) - 26(4)$$

$$\Rightarrow \boxed{15^{-1} = 7}$$

⑦ $26 = 17(1) + 9$

$$17 = 9(1) + 8$$

$$9 = 8(1) + 1$$

$$1 = 9 - 8(1)$$

$$1 = 9 - 17 + 9$$

$$1 = 2(26 - 17) - 17$$

$$1 = 26(2) - 17(3)$$

$$\Rightarrow \boxed{17^{-1} = 23}$$

$$\Rightarrow \boxed{23^{-1} = 17}$$

⑧ $26 = 19(1) + 7$

$$19 = 7(2) + 5$$

$$7 = 5(1) + 2$$

$$5 = 2(2) + 1$$

$$1 = 5 - 2(2)$$

$$1 = 5 - 7(2) + 5(2)$$

$$1 = 19(3) - 7(6)$$

$$1 = 19(3) - 6(26 - 19)$$

$$1 = 19(19) - 6(26)$$

$$\boxed{19^{-1} = 11}$$

⑨ $21^{-1} = 5$

from ②

⑩

$$26 = 5(5) + 1$$

$$1 = 26 - 5(5)$$

$$1 = 26 - 25$$

② Affine Cipher

Multiplier = 5 add = 8

$$\text{AFFINE} \quad E = PK_1 + K_2 \pmod{26}$$

$$A \quad 5(0) + 8 \pmod{26} = 8 \quad I$$

$$F \quad (5 \times 5 + 8) \pmod{26} = 23 \quad H$$

$$F \quad (5 \times 6 + 8) \pmod{26} = 7 \quad H$$

$$I \quad (5 \times 8 + 8) \pmod{26} = 22 \quad W$$

$$N \quad (5 \times 13 + 8) \pmod{26} = 21 \quad V$$

$$E \quad (5 \times 4 + 8) \pmod{26} = 2 \quad C$$

AFFINE = I H H W V C

③ HI = C a = 11 b = 5

$$E = (P \times a + b) \pmod{26}$$

$$H = (11 \times 7 + 5) \pmod{26} = 4 = E \quad HI = EP$$

$$I = (11 \times 8 + 5) \pmod{26} = 15 = P$$

④ VCR $E = 9P + 2 \pmod{26}$

$$P = 9^{-1}(C - 2) \pmod{26} \Rightarrow P = 3(C - 2) \pmod{26}$$

$$26 = 9(2) + 8 \quad U = 20 = 3(20 - 2) \pmod{26} = 2 = C$$

$$9 = 8(1) + 1 \quad C = 2 = 3(2 - 2) \pmod{26} = 0 = A$$

$$1 = 9 - 8 \quad R = 17 = 3(17 - 2) \pmod{26} = 19 = T$$

$$1 = 9 - (26 - 9(2))$$

$$1 = 3(9) - 26$$

$$\boxed{9^{-1} = 3}$$

VCR → CAT

⑤ Decrypt : K D Y X J U W D E X Z Y

$$a = 3 \quad b = 4$$

$$C = (3P + 4) \pmod{26}$$

$$P = 3^{-1}(C - 4) \pmod{26} = 9(C - 4) \pmod{26}$$

$$\underline{\underline{3^{-1} = 9}}$$

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$$\begin{aligned}
 K &= 9(10-4) \bmod 26 = 2 = C \\
 D &= 9(8-4) \bmod 26 = 17 = R \\
 Y &= 9(24-4) \bmod 26 = 24 = Y \\
 X &= 9(23-4) \bmod 26 = 15 = P \\
 J &= 9(9-4) \bmod 26 = 19 = T \\
 U &= 9(20-4) \bmod 26 = 14 = O \\
 W &= 9(22-4) \bmod 26 = 6 = G \\
 D &= 9(3-4) \bmod 26 = -9 \Rightarrow 17 = R \\
 E &= 9(4-4) = 0 = A \\
 X &= 9(23-4) \bmod 26 = 15 = P \\
 Y &= 9(25-4) \bmod 26 = 7 = H \\
 &= 24 = Y
 \end{aligned}$$

$\Rightarrow$ CRYPTOGRAPHY

⑥ $k_1 = 13$, $k_2 = 4$
 $C = (13P + 4) \bmod 26$
 $(13, 26)$ is not Co-prime $\Rightarrow k_1$ is not valid.

Encryption

$$\begin{aligned}
 I &= 13(8) + 4 \bmod 26 = 4 = E \\
 N &= (13 \times 13 + 4) \bmod 26 = 17 = R \\
 P &= (13 \times 15 + 4) \bmod 26 = 17 = R \\
 U &= (13 \times 20 + 4) \bmod 26 = 4 = E \\
 T &= (13 \times 19 + 4) \bmod 26 = 17 = R
 \end{aligned}$$

INPUT = ERROR

$$\begin{aligned}
 A &\rightarrow 0 & (13 \times 0 + 4) \bmod 26 = 4 = E & \Rightarrow \text{ERROR} \\
 L &\rightarrow 11 & (13 \times 11 + 4) \bmod 26 = 17 = R \\
 T &\rightarrow 19 & (13 \times 19 + 4) \bmod 26 = 17 = R \\
 E &\rightarrow 4 & (13 \times 4 + 4) \bmod 26 = 4 = E \\
 R &\rightarrow 17 & (13 \times 17 + 4) \bmod 26 = 17 = R
 \end{aligned}$$

Different Plain text produces same Cipher text. So decryption is not uniquely possible

⑦ $K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \bmod 26$: Inverse?

$$|K| = 3 \times 7 - 5 \times 2 = 21 - 10 = 11$$

$$K^{-1} = \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix}$$

$$\text{Inverse} = 11 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26}$$

$$11 \begin{bmatrix} 7 & -2 \\ -5 & 3 \end{bmatrix} \pmod{26} = \begin{bmatrix} 133 & -18 \\ -95 & 57 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 3 & 14 \\ 9 & 5 \end{bmatrix}$$

$$\textcircled{8} \begin{bmatrix} 11 & 13 \\ 2 & 3 \end{bmatrix}$$

$$K^{-1} = \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix}$$

$$|K| = 33 - 26 = 7$$

$$7^{-1} = 15$$

$$\text{Inverse} = 15 \begin{bmatrix} 3 & -13 \\ -2 & 11 \end{bmatrix} \pmod{26}$$

$$= \begin{bmatrix} 45 & -195 \\ -30 & 165 \end{bmatrix} \pmod{26} = \begin{bmatrix} 19 & 13 \\ 22 & 9 \end{bmatrix}$$

$$\textcircled{9} \begin{bmatrix} 2 & 14 \\ 13 & 21 \end{bmatrix} \text{ Key Valid?}$$

$$|K| = 2 \times 21 - 13 \times 14 = 42 - 182 = -140 \pmod{26} = 16$$

$$16, 26 \quad \text{gcd} = 2$$

$\therefore 16$ cannot be a valid key.

$$19^{-1} \bmod 26 = 11$$

SOLVED

$$K = \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 18 \\ 14 \end{bmatrix}$$

$$\begin{bmatrix} 266 \\ 280 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 20 \end{bmatrix} = \begin{matrix} K \\ U \end{matrix}$$

LV

$$\begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 201 \\ 308 \end{bmatrix} \bmod 26 = \begin{bmatrix} 19 \\ 22 \end{bmatrix} = \begin{matrix} T \\ W \end{matrix}$$

$$ED \quad \begin{bmatrix} 3 & 8 \\ 7 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 36 \\ 61 \end{bmatrix} \bmod 26 = \begin{bmatrix} 10 \\ 9 \end{bmatrix} = \begin{matrix} K \\ J \end{matrix}$$

$$\Rightarrow KUTWKJ$$